

Newton's law of motion

Introduction to Newton's Law of Motion:

What is Dynamics: Dynamics is the branch of mechanics that deals with the motion of bodies under the action of forces.

Why are Newton's laws important:

They describe the relationship between motion and force.

They help in understanding not just in the motion of planets, but also daily life - phenomena like walking, driving and flying.

Examples of Dynamics:



What have we studied till now in class 9th:

• Newton's 1st Law of Motion: A body at rest remains at rest or a body in motion remains in motion until an external force

Newton's 2nd Law of Motion: Force acting on any body is directly proportional to the acceleration of the body.

$$F = Ma$$

Force → Cause
Acceleration → effect

• Newton's 3rd Law of Motion: This law states that for every action, there is equal and opposite reaction.

Aristotle's Fallacy: → Incorrect theory

• What did Aristotle believe:

- Aristotle believed that a force is necessary to keep a body in motion.
- According to him, the natural state of an object is rest, and any motion - require a continuous external force.

• Why is this incorrect? (The Fallacy):

- This view was disproved by Galileo and later formulated correctly by Newton.



- In the absence of friction, a body in motion continues to move without any force. Hence, continuous force is not needed for uniform motion.

The Law of Inertia: Inertia is the tendency of a body to oppose the change in its state
↓
(Rest or Motion)

Inertia of a body $\propto$ Mass of the body

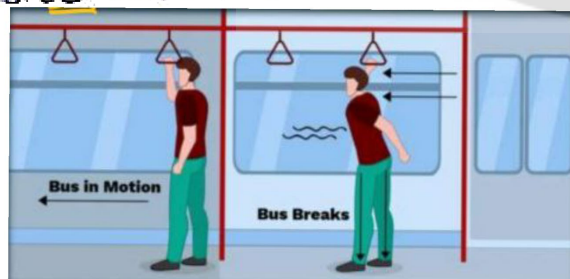
Types of Inertia:

- i) Inertia of Rest.
- ii) Inertia of Motion.
- iii) Inertia of Direction.

Inertia of Rest: If a body is at rest, then it has a tendency to remain at rest for some time, even after an external force is applied to it. This tendency is known as Inertia of Rest.



Inertia of Motion: A body in motion has the tendency to remain in motion for some time even after external force is applied to it. This tendency is called Inertia of Motion.



Inertia of Direction: A body moving in a particular direction has a tendency to stay in that direction for some time even after an external force is applied on it. The tendency is known as Inertia of Direction.



Newton's First Law of Inertia:

A body remains in its state of rest or uniform motion in a straight line unless acted upon by an external force.

$$\begin{aligned}
 [F=0] & \quad [a=0] \rightarrow a = \frac{v-u}{t} \rightarrow a = \frac{\Delta v}{\Delta t} \\
 \text{Guru} & \quad \text{Chela} \quad 0 = \frac{v-u}{t} \\
 & \quad \quad \quad = v-u=0 \rightarrow \boxed{v=u} = \boxed{\Delta v=0}
 \end{aligned}$$

Momentum $\rightarrow$ SI unit of momentum $\rightarrow \text{kgms}^{-1}$

Momentum of a body is defined as the product of its mass and velocity.

Momentum = Mass $\times$ velocity

$$[\vec{p} = m \times \vec{v}]$$

Nature:

- It is a vector quantity
- Direction of Momentum is same as the velocity

Physical-Significance:

- More is the mass and the velocity of the moving object, more is the force required to stop its motion

Newton's Second Law of Motion:

Also known as **[The law of Momentum]**

- The rate of change of momentum of an object is directly proportional to the applied force and takes place in the direction of the applied force.

$$[\vec{F} \propto \frac{\Delta \vec{p}}{\Delta t}] \rightarrow \text{we are going to derive the expression / formula of force from the relation.}$$

Mathematical form of Newton's Second Law:

Given: mass of the body $\rightarrow m$
 Initial Velocity $\rightarrow u$
 Final velocity $\rightarrow v$
 Momentum $\rightarrow p$
 time $\rightarrow t$

We know that,

$$\vec{p} = m\vec{u}$$

Initial Momentum -

$$\vec{p}_i = m\vec{u} \quad \text{--- (1)}$$

Final Momentum -

$$\vec{p}_f = m\vec{v} \quad \text{--- (2)}$$

Change in Momentum ---

$$\Delta \vec{p} = \vec{p}_f - \vec{p}_i$$

$$\Rightarrow \Delta \vec{p} = m\vec{v} - m\vec{u} \quad [\text{from eqn (1) and eqn (2)}]$$

$$\Rightarrow \Delta \vec{p} = m(\vec{v} - \vec{u}) \quad \text{--- (3)}$$

From Newton's 2nd law of motion ---

$$\vec{F} \propto \frac{\Delta \vec{p}}{\Delta t} \Rightarrow \vec{F} = k \frac{\Delta \vec{p}}{\Delta t} \quad \left\{ \begin{array}{l} \text{where } k \text{ is} \\ \text{a constant} \\ \text{and } k=1 \end{array} \right.$$

$$\Rightarrow \vec{F} = \frac{m(\vec{v} - \vec{u})}{\Delta t} \quad [\text{from eqn (3)}]$$

$\Delta \vec{v}$

$$\Rightarrow \vec{F} = m \frac{\Delta \vec{v}}{\Delta t} \rightarrow \vec{a} = \boxed{\vec{F} = m\vec{a}}$$

The SI-unit of Force: $\vec{F} = m\vec{a}$

Force = 'mass' $\times$ Acceleration

$[kgms^{-2}]$ $[kg]$ $[ms^{-2}]$

Newton $\rightarrow$ SI unit of Force
 Denoted by 'N'.

Applications of Newton's Second Law:

Law:

- **Kicking a Football:** Greater force $\rightarrow$ more acceleration.
- **Cricket fielder:** Lowers hands to catch a ball: increase time $\rightarrow$ reduces Force.
- **Rocket Propulsion:** Mass of gases pushed backward produces Forward Force.
- **Martial arts breaking bricks:** Large Force applied over short time.



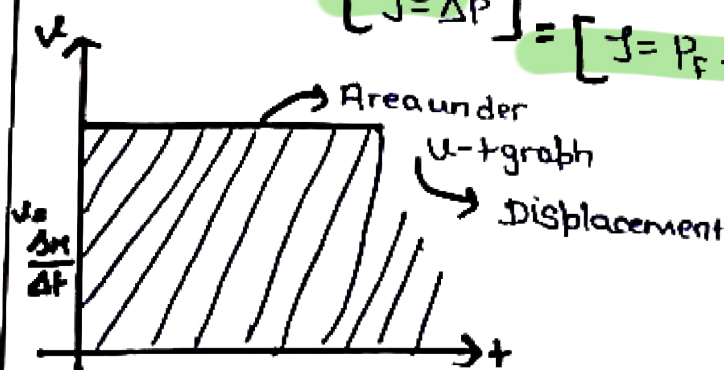
Impulse: Impulse is the total effect of a force acting on a body for a short time.
 [Impulse = $F \times \Delta t$]

Graphical Meaning: Area under $F-t$ graph

Units: N.s or $kg \cdot m/s$ (Same as Momentum)
 Impulse = Change in Momentum.

$$[J = F \times \Delta t] = \Delta p$$

$$[J = \Delta p] = [J = p_f - p_i]$$

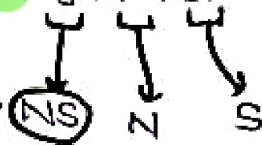


Graphical Meaning: Area under F-t graph

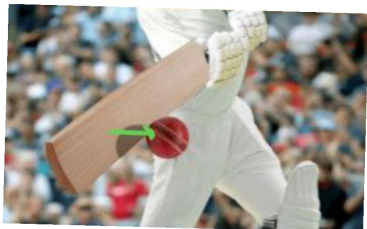
Unit of Impulse:

Impulse

$$J = F \times \Delta t$$

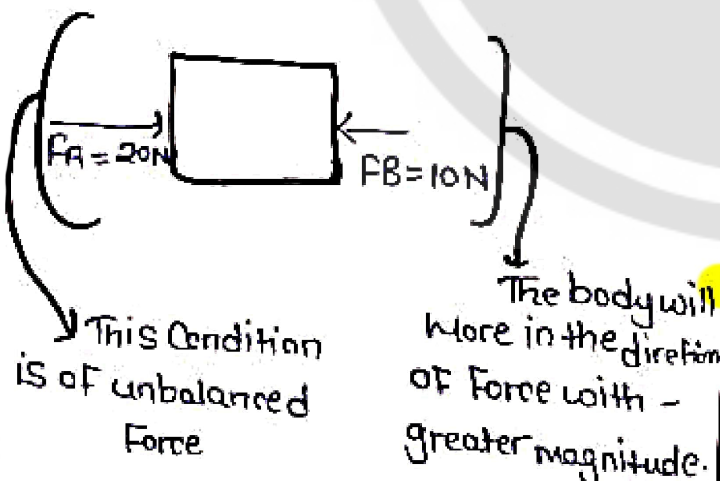
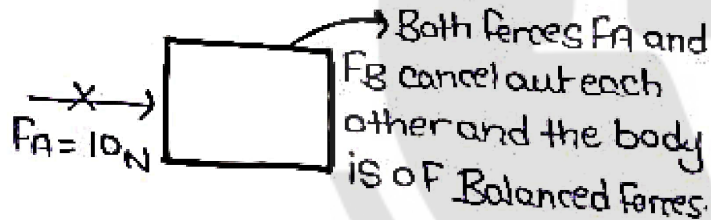


Example:



Balanced and Unbalanced Forces:

All the forces must be applied on a single body



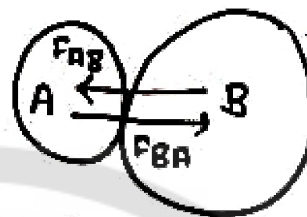
Newton's Third Law of Motion:

- For every action, there is an equal and opposite reaction.

- Forces occur in pairs, equal in magnitude, opposite in direction, act on different bodies.

Key points:

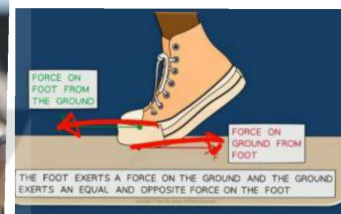
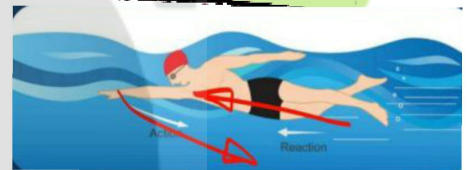
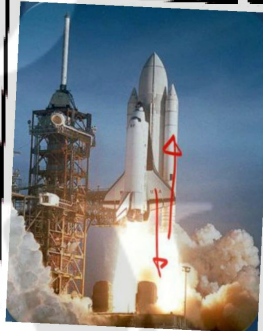
- Action-Reaction forces never cancel each other (different bodies).
- Work in pairs at the same instant.



Both the forces are equal and opposite.

$$[F_{AB} = -F_{BA}]$$

Examples of Newton's Third Law:



Law of Conservation of Momentum:

In the absence of an external force, total momentum of a system remains constant.

Equilibrium of a particle:

- A particle is in equilibrium if the net force on it is zero.
- Condition for equilibrium: Vector sum of all forces = 0;

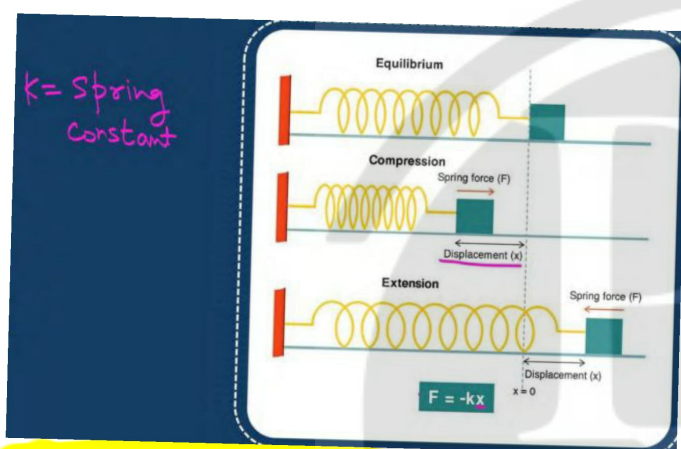
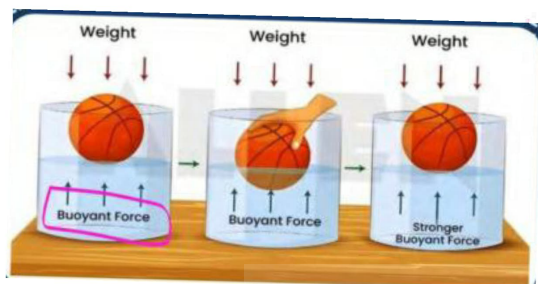
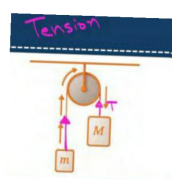
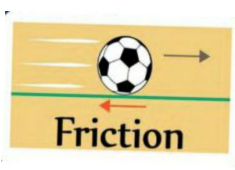
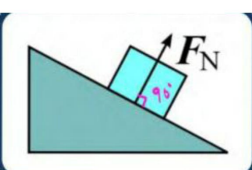
$$[\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots = 0]$$

Types:

- Static Equilibrium: Particle at rest, forces balanced.

Dynamic Equilibrium: Particle moving with constant velocity, net force zero.

Common Forces in Mechanics:

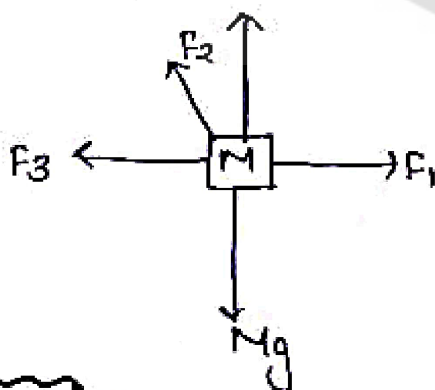


Solving Problems in Mechanics:

Concept of FBD

Facebook Dp

Free body Diagram



Friction (F)

- Whenever two surfaces are in contact and one tries to move over the other, there is always a force that resists this relative motion.

This opposing force is called Friction.

- It always acts along the contact surface and opposite to motion or tendency of motion.

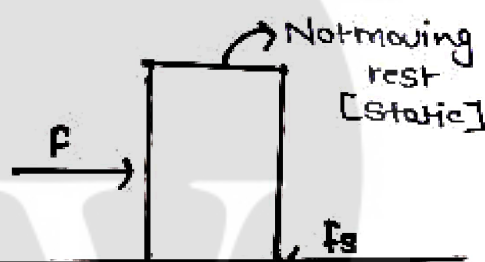
Friction arises mainly due to two reasons:

1. Microscopic irregularities of surface:

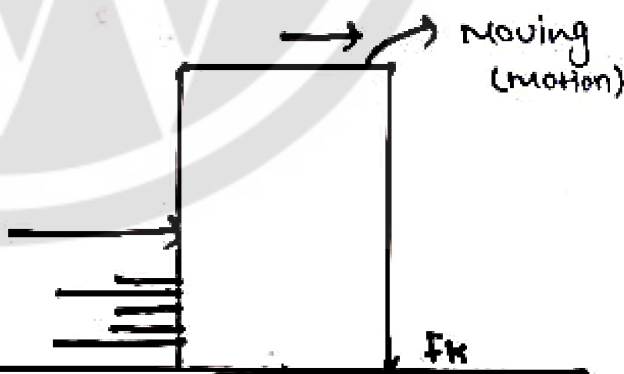
- Even smooth surfaces have tiny bumps and hollows.
- These interlock and resist motion.

2. Molecular forces of Adhesion:

- At very close contact, molecules of the two surfaces exert weak adhesive forces further resisting motion.



Static Friction



Kinetic Friction

Static Friction (F_s): Frictional force acting on the body when it is static or at rest - irrespective of the ~~forces~~ forces applied on it, is called static friction.

$$[f_s \leq \mu_s N]$$

Kinetic Friction (f_k)

- Frictional Force acting on the body when its in motion is called the kinetic called the kinetic Friction.

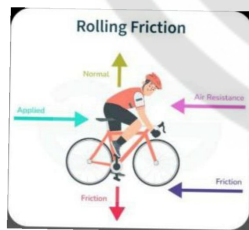
$$[f_k = \mu_k N]$$

↓
Coefficient of kinetic Friction.

- Determine the maximum acceleration of the train in which a box lying on its floor will remain stationary. given that the co-efficient of static friction between the box and the train floor is 0.16.

Rolling Friction:

- Opposes the motion of rolling objects like wheels, balls, cylinders.
- Very small compared to sliding friction.
- That's why vehicles use wheels, and heavy luggage is moved using rollers.



Circular Motion: $\tan \theta = \frac{v^2}{Rg}$

- θ = Angle of elevation of the road
- v = Velocity of the body.
- R = Radius of the circular turn.

